Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions

Molena Huynh[1,*]

[1] *North Carolina State University, Raleigh, North Carolina 27695, USA*

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Classical optimization can dominate the measurement cost of a variational quantum algorithm, yet learned optimizer routing may be less reliable than a fixed schedule. We cast graph-conditioned trust regions as a target-free, fixed-budget portfolio for depth-two MaxCut QAOA in which nine arms share a 32-call ceiling. A ridge model predicts each arm's anytime regret, a known graph family supplies a strong fallback, and a one-sided residual order statistic abstains whenever the predicted improvement is not certified. After a 64-graph development audit, the selector is fit on 48 graphs, calibrated on 48 disjoint graphs, and locked before a single evaluation on 160 instances. Gated deployment attains mean regret 0.16930 against 0.17207 for the fallback, but accepts only 12 routes, four of them harmful. One-sided empirical coverage is 0.88125, below the frozen 0.90 requirement, so the reliability claim fails; fixed family quotas also preclude a pooled exchangeability certificate. The contribution is a reproducible exact-statevector failure analysis, not evidence of hardware, scaling, or quantum advantage.

I. INTRODUCTION

A variational quantum algorithm pairs a parameterized circuit with a classical optimizer [1, 2]. Its near-term appeal comes with a stubborn cost problem: each black-box objective query can demand many circuit preparations and measurements [3, 4], so the optimizer belongs to the physical resource model rather than to interchangeable software [5].

The quantum approximate optimization algorithm (QAOA) makes the coupling concrete by alternating cost and mixer unitaries before optimizing a small angle vector [6, 7], with combinatorial objectives encoded as Ising Hamiltonians [8]. Its empirical behavior turns on graph structure, depth, initialization, and search budget [9]. Angle concentration can make fixed schedules competitive on typical instances [10, 11], warm starts can inject classical solutions [12], and transferred parameters can exploit similarity across weighted graphs [13].

Graph-conditioned optimization tries to use this shared structure before or during search: a predictor can propose angles, a local covariance, a trust region, or an optimizer choice. Learned graph initializers demonstrate the opportunity [14], and learned variational optimizers show that trajectories themselves can be predicted [15]. The statistical question is rarely stated: when should a learned recommendation displace a strong, cheap control?

That question cannot be settled by average terminal quality. Tuning and selection on the same benchmark used for evaluation introduce substantial optimism [16], a router can raise a pooled mean by helping one family while harming another, and a target-based stopping rule can depend on an oracle unavailable in deployment. The classical algorithm-selection problem already separates instance features, candidate algorithms, and performance measures [17]; the quantum setting merely adds an unusually expensive response surface.

We therefore replace privileged target hitting with a fixed-query anytime metric. Every arm receives 32 objective calls, and utility is the average normalized regret over the checkpoints $\{1, 2, 4, 8, 16, 32\}$. A relabeling-invariant structural model predicts each arm's utility, the strongest development arm within the known graph family is the fallback, and routing is accepted only when an upper residual order statistic indicates that the selected arm should beat that fallback. The abstention rule borrows split-conformal reasoning [18, 19], whose finite-sample guarantee needs exchangeability; because our confirmatory design uses fixed family quotas with sequential non-isomorphism filtering, we report the frozen rule together with the limits of its coverage interpretation [20].

Figure 1 fixes the evidence chronology. Development fits the selector and family fallback, calibration fixes one residual order statistic, the graph list, code, configuration, and decision are hashed, and the audit is evaluated once. This chronology matters as much as the model, because unrecorded iteration on audit outcomes would erase the meaning of a confirmatory result. The final decision is negative: the deployed policy slightly improves mean anytime regret, accepts only 12 of 160 routes, harms four accepted instances, and misses the frozen coverage threshold. A favorable mean coexisting with a failed reliability criterion is the distinction this paper is built to preserve.

* molena.huynh@jmp.com

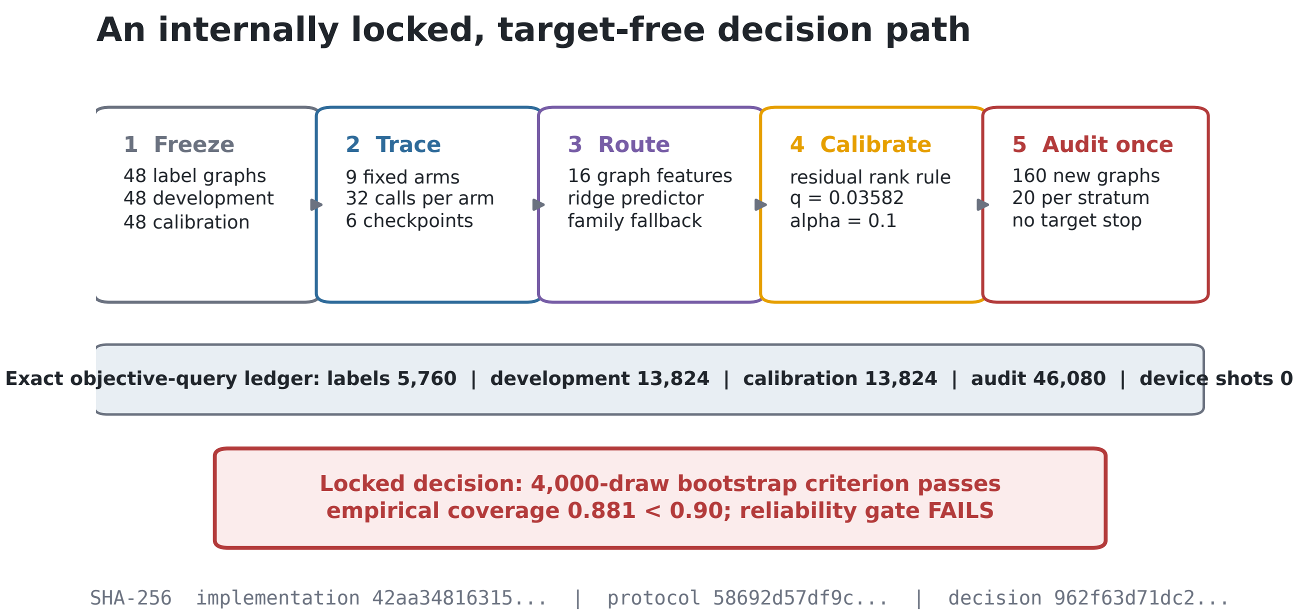

FIG. 1. **Internally locked fixed-budget protocol.** Development data fit the structural selector and family fallback; calibration fixes a one-sided residual order statistic. The graph list, implementation, configuration, and decision are hashed before the 160-graph audit is evaluated.

II. BACKGROUND

A. Quantum Approximate Optimization

For an unweighted graph $G = (V, E)$, define

$$C_G = \sum_{(u,v)\in E} \frac{I - Z_u Z_v}{2}, \qquad B = \sum_{v\in V} X_v. \tag{1}$$

A depth-p QAOA state is

$$|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = \prod_{j=p}^{1} e^{-i\beta_j B} e^{-i\gamma_j C_G} |+\rangle^{\otimes |V|}, \tag{2}$$

with objective $F_G(\boldsymbol{\theta}) = \langle \boldsymbol{\gamma}, \boldsymbol{\beta} | C_G | \boldsymbol{\gamma}, \boldsymbol{\beta}\rangle$. Analytic parameter-shift rules evaluate gradients on compatible hardware [21, 22], whereas stochastic estimators trade coordinate resolution for fewer objective calls [23].

Gradient methods do not dissolve trainability concerns. Random or highly expressive circuits can exhibit barren plateaus [24]; expressibility is tied to gradient concentration [25]; and control-theoretic tools sharpen the diagnosis [26]. Gradient-free search likewise degrades in flat landscapes [27]. We therefore compare complete fixed-budget trajectories instead of presuming one access model is uniformly best.

B. Fixed-Budget Black-Box Optimization

At the checkpoints $\mathcal{T} = \{1, 2, 4, 8, 16, 32\}$, arm a on instance i has post hoc normalized regret

$$r_{ia}(t) = 1 - \frac{\max_{s\le t} F_{G_i}(\boldsymbol{\theta}_{ias})}{C_{\max}(G_i)} \tag{3}$$

and utility

$$u_{ia} = \mathrm{AURC}_{ia} = \frac{1}{|\mathcal{T}|} \sum_{t\in\mathcal{T}} r_{ia}(t), \tag{4}$$

where lower is better. Exact MaxCut $C_{\max}$ enters only after optimization; no arm ever observes a target or a target-hitting event.

The portfolio spans fixed schedules, transferred angles, multistart, SPSA, and the graph-conditioned trust-region arm, a heterogeneity that reflects the fact that optimizer rankings depend on the problem and budget [28]. Mesh-adaptive direct search gives a principled derivative-free comparator [29], BOBYQA maintains a bounded quadratic model [30], and covariance-matrix adaptation learns a search distribution [31]. We keep the simpler arms because a portfolio is only meaningful relative to its actual controls, and a ramped fixed schedule is a strong low-complexity baseline [32].

C. Selective Routing and Risk

Let $x(G) \in \mathbb{R}^{16}$ be a relabeling-invariant graph summary and $\widehat{u}_a(x)$ the predicted utility of arm a. The proposed arm is

$$s(x) = \arg\min_a \widehat{u}_a(x). \tag{5}$$

For family label f, the fallback $b(f)$ is the development arm of minimum family-mean utility. Define the realized selected-minus-fallback regret and its prediction residual

$$\Delta = u_s - u_b, \qquad R = \Delta - (\widehat{u}_s - \widehat{u}_b). \tag{6}$$

Given a calibration upper order statistic q, routing is accepted when

$$\widehat{u}_s - \widehat{u}_b + q < 0, \tag{7}$$

and otherwise abstains to $b(f)$. This is selective prediction, in which coverage, acceptance, joint harm, and conditional harm are different quantities. Conformalized quantile regression can adapt interval width to covariates [33], and exact binomial limits can summarize observed event rates [34], but neither licenses a coverage claim without a sampling model.

III. RELATED WORK

QAOA initialization exploits concentration, interpolation, and problem structure [9, 10], and learned graph initializers extend the idea through message passing [14]. Weighted-MaxCut parameter transfer targets related objectives [13], while ramp schedules supply a strong low-complexity comparator [32]. Our portfolio treats these ideas as arms or controls rather than assuming one initialization dominates.

Graph convolutional networks use normalized neighborhood aggregation [35], graph attention learns neighbor weights [36], and graph-isomorphism networks characterize a powerful message-passing class [37]. We deliberately use 16 hand-computed invariant summaries with ridge regression; the low-capacity choice reduces model flexibility and makes every routing score inspectable.

Bayesian optimization is another route to expensive black-box search. Efficient global optimization introduced improvement-based acquisition [38], practical formulations popularized Gaussian-process tuning [39], a broad review organized surrogate and acquisition choices [40], and differentiable Monte Carlo acquisitions are available in modern toolkits [41]. These may be useful arms, yet fitting a surrogate does not make its objective evaluations free.

Algorithm configuration learns settings across tasks: sequential model-based configuration proposes new settings from prior runs [42], and SMAC3 exposes a reproducible modern package [43]. The portfolio view also touches active subspaces [44], sliced inverse regression [45], and regression graphics [46], though our predictor reduces neither the circuit parameter dimension nor the feature dimension adaptively. Randomized linear algebra would matter for future low-rank structural models: randomized range finding and blocked QB give scalable, adaptive-rank approximation [47, 48], sketching offers streaming alternatives [49, 50], and matrix concentration supplies probabilistic tools [51]; none is needed for the released 16-feature ridge fit. Finally, statistical safeguards address distinct risks: bootstrap resampling estimates sampling variation [52], multiple-testing procedures control false discoveries [53], the Neyman–Pearson framework separates size from power [54], and cross-data-set comparison must occur at the data-set level [55].

IV. FAILURE MODES OF STRUCTURAL OPTIMIZER ROUTING

A. Selection Bias and Distribution Shift

Choosing a route because it is best on development data is not a distribution-free argument. Even a perfect empirical ranking can invert under task shift, a basic distinction in algorithm selection and honest model assessment [16, 17].

Theorem IV.1 (Near-maximal excess loss after routing shift)**.** *For every $\varepsilon \in (0,1)$ there exist two optimizer arms a, b, losses in $[0,1]$, a development distribution P_0, and a deployment distribution P_1 such that a is the unique risk minimizer under P_0, b is the unique risk minimizer under P_1, and routing unconditionally to a has deployment excess risk $1-\varepsilon$.*

Appendix A 1 gives a complete construction. The theorem is elementary by design: no model complexity is required to produce a failure, which is exactly why graph splits, family strata, and chronology must be declared before a learned router is interpreted. The project's first version optimized against an offline target and compared a learned trust-region policy mainly with random restart; stronger fixed schedules were cheaper, and the target was unavailable at deployment. The present portfolio keeps that history as a legacy diagnostic and does not merge it with the target-free confirmatory result.

The one-shot outcome and its family concentration are displayed in Fig. 2.

B. Coverage and Conditional Harm

Let A be the event that the selector accepts a route and $H = \{\Delta > 0\}$ the event that the selected arm is worse than the fallback.

Proposition IV.2 (Harm-rate decomposition)**.** *If $\mathbb{P}(A) > 0$, then*

$$\mathbb{P}(A \cap H) = \mathbb{P}(A)\,\mathbb{P}(H \mid A), \tag{8}$$

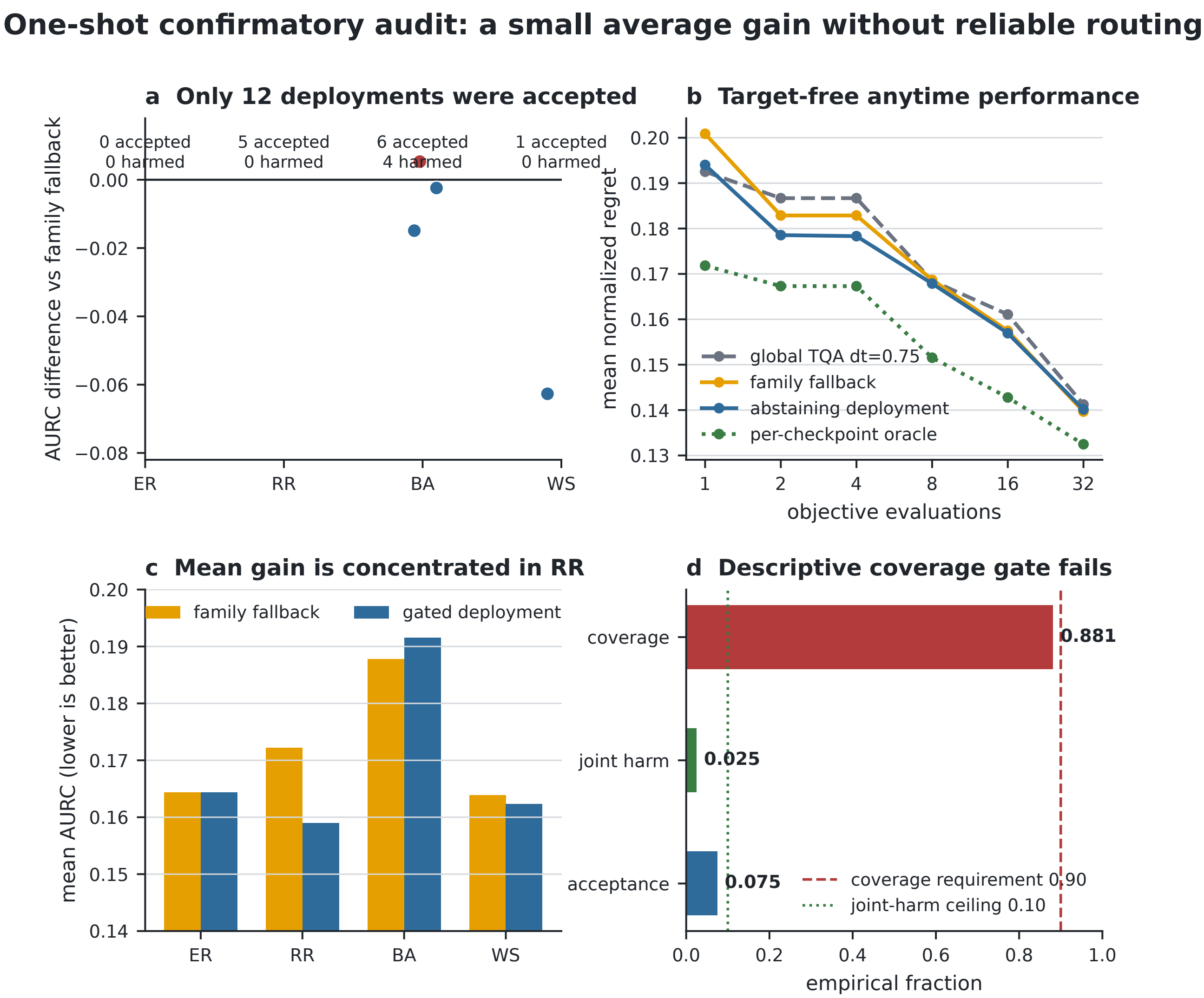

FIG. 2. **One-shot confirmatory audit.** The mean gain is small and concentrated by family. Twelve routes are accepted, four are harmful, and descriptive one-sided coverage misses the frozen requirement.

so controlling joint harm alone does not control conditional harm when acceptance can be small.

The proof and an explicit counterexample are in Appendix A 2. In the audit, $\mathbb{P}_n(A) = 12/160 = 0.075$, $\mathbb{P}_n(A \cap H) = 4/160 = 0.0250$, and $\mathbb{P}_n(H \mid A) = 4/12 = 0.333$. The joint rate is small because acceptance is small, not because accepted routes are uniformly safe. Marginal coverage is also distinct from selective harm: an upper residual bound can cover Δ while acceptance concentrates errors near the decision boundary, and conversely a policy can have acceptable joint harm while missing a nominal residual-coverage target. All four quantities must be reported, because marginal calibration and conditional selection answer different questions [18, 19].

C. Acceptance, Exceedance, and Non-Certification

Write $\widehat{d} = \widehat{u}_s - \widehat{u}_b$, $R = \Delta - \widehat{d}$, $A = \{\widehat{d} + q < 0\}$, and $H = \{\Delta > 0\}$. The acceptance rule has a deterministic implication that needs no sampling model.

Proposition IV.3 (Acceptance–exceedance containment)**.** *For every joint distribution of $(\Delta, \widehat{d})$ and every $q \in \mathbb{R}$,*

$$A \cap H \subseteq \{R > q\}. \tag{9}$$

Consequently, for any $\alpha \in [0, 1]$, if $\mathbb{P}(R \leq q) \geq 1 - \alpha$ then $\mathbb{P}(A \cap H) \leq \alpha$; if additionally $\mathbb{P}(A) > 0$ then $\mathbb{P}(H \mid A) \leq \alpha/\mathbb{P}(A)$.

Appendix A 3 proves every implication. The result clarifies both the appeal and the limit of residual calibration:

idealized marginal residual coverage controls joint harm, whereas the conditional bound deteriorates as acceptance becomes rare. More importantly, the locked fixed-quota, sequentially filtered audit does not establish the pooled exchangeability needed to infer the premise from the order statistic, so Proposition IV.3 is an algebraic reading of the gate rather than a certificate for the released audit [20]. Balancing an audit across families changes the sampling law: even if the final order is randomized, family labels are dependent once their totals are fixed, so comparisons must respect the stratum as the sampling unit [52, 55].

Proposition IV.4 (Fixed quotas are not pooled iid sampling)**.** *Consider an audit of size n with exactly n_f graphs from family f, where $0 < n_f < n$, and let the order of the n graphs be uniformly randomized. For distinct positions $i \neq j$, with $Z_i = \mathbf{1}\{\text{position } i \text{ has family } f\}$ and $p_f = n_f/n$,*

$$\mathrm{Cov}(Z_i, Z_j) = -\frac{p_f(1-p_f)}{n-1} < 0. \tag{10}$$

Hence the pooled family labels are exchangeable but not independent, and the audit is not an iid sample from the mixture with family mass p_f.

Appendix A 4 proves the covariance identity by direct counting. The proposition does not invalidate stratum-specific description or a design-based analysis; it limits only an iid interpretation that ignores the fixed quotas and sequential filtering.

Figure 3 contrasts the frozen development and confirmatory decisions without reusing the audit split for tuning.

V. RISK-CONTROLLED GRAPH-CONDITIONED TRUST REGIONS

A. Target-Free Portfolio Utility

Every arm receives the same 32-call counter; seed evaluations, restarts, and all intermediate queries count against the ceiling. The utility in Eq. (4) rewards useful values early without granting any optimizer access to $C_{\max}$, so the metric is target free for control even though exact MaxCut remains a small-instance analysis oracle. The nine arms are a training-set concentration vector, four midpoint linear schedules, one-nearest-neighbor transfer, four-start random multistart, SPSA, and the legacy graph-conditioned Gaussian trust-region policy. A fixed schedule can be competitive because QAOA angles concentrate [11], so the portfolio retains simple arms instead of comparing only learned policies.

B. Structural Prediction and Family Fallback

The structural vector contains size, edges per vertex, density, seven degree summaries, mean clustering, transitivity, triangles per vertex, and three normalized-Laplacian summaries. A multi-output ridge model predicts each arm's utility,

$$\widehat{\boldsymbol{u}}(x) = \boldsymbol{a} + W\widetilde{x}, \qquad \lambda_{\text{ridge}} = 1, \tag{11}$$

where normalization uses development means and scales only. The known synthetic family label defines a strong deployable fallback, a choice deliberately favorable to the control: the router must improve on what can be selected without instance-level prediction. General algorithm-selection theory warns that larger portfolios raise both opportunity and overfitting risk [16, 17].

C. Residual Order-Statistic Abstention

Fix $\alpha \in (0,1)$, and suppose calibration residuals $R_1, \ldots, R_n$ and a future residual R_{n+1} are exchangeable and almost surely distinct. Let $k = \lceil (n+1)(1-\alpha) \rceil$ and let q be the kth calibration order statistic, with $q = +\infty$ when $k > n$.

Theorem V.1 (Split order-statistic coverage)**.** *Under these assumptions,*

$$\mathbb{P}(R_{n+1} \le q) = \frac{k}{n+1} \ge 1 - \alpha. \tag{12}$$

Appendix A 5 proves the result by rank symmetry. Ties are handled conservatively or by randomized ranks. The theorem explains the calibration rule but does not assert that the quota-filtered audit residual is exchangeable with the pooled calibration residuals; here $n = 48$ and $\alpha = 0.10$ give the rank-45 upper statistic.

D. Locked Confirmatory Decision

The configuration, graph identities, angle labels, implementation, non-audit traces, learned state, calibration statistic, and decision are hashed before the audit is evaluated, and the evidence directory refuses a second audit run; the generator revalidates these hashes before producing manuscript displays. This scheme is not an external preregistration and does not remove every researcher degree of freedom. The hashes establish internal consistency with the declared execution order and the output lock refuses an overwrite, but they are not an independent timestamp and cannot exclude unrecorded alternative runs. Reproducibility here concerns both computational determinism and inferential chronology.

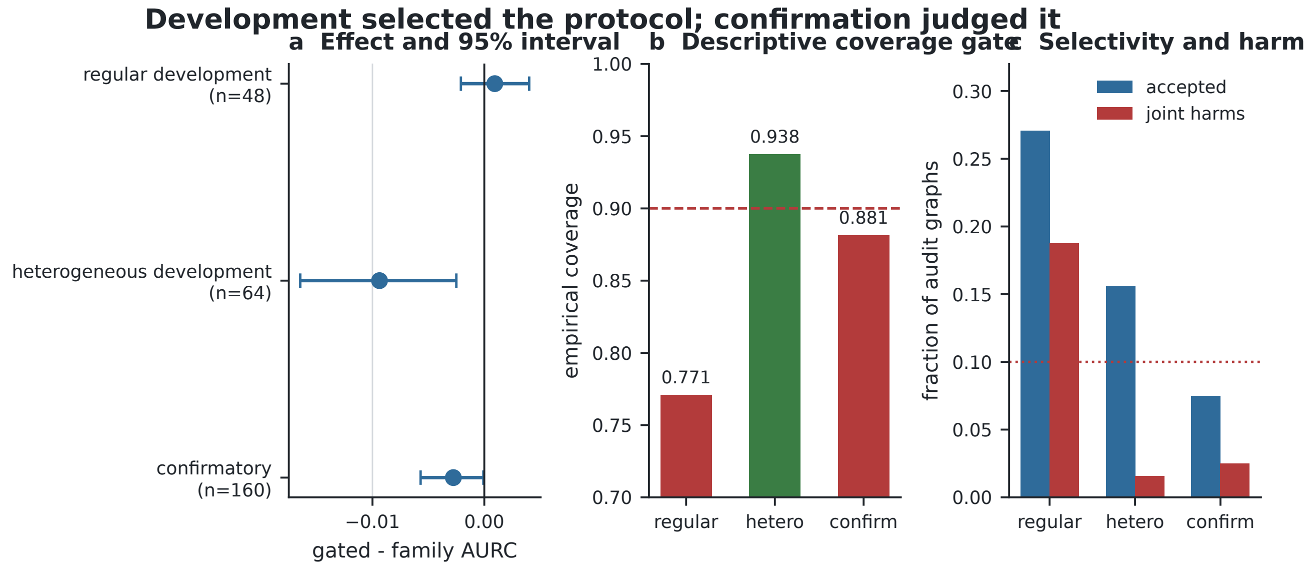

FIG. 3. **Development-to-confirmation progression.** The heterogeneous development split satisfies the frozen gate, while the confirmatory split misses its empirical-coverage component. Audit outcomes were not used to retune the released decision.

VI. EXPERIMENTS

The locked benchmark asks whether a graph-conditioned selector lowers fixed-budget anytime regret relative to a strong graph-family fallback and, separately, whether accepted routes satisfy the declared reliability gate. The population is a controlled Cartesian design over Erdős–Rényi, random-regular, Barabási–Albert, and Watts–Strogatz generators with $n \in \{12, 14\}$ and QAOA depth two. Graph identities separate 48 angle-label instances, 48 development instances, 48 calibration instances, and 160 untouched audit instances. The graph is the statistical sampling unit; family and size define the eight fixed bootstrap strata.

Nine frozen optimizer arms receive the same 32 exact-objective-call ceiling: concentration, four transferred fixed schedules, nearest-neighbor transfer, random multistart, SPSA, and legacy GCTR. Development fixes the ridge router and family fallback, calibration fixes the one-sided residual order statistic, and the audit is scored once without refitting. The primary endpoint is selected-minus-family-fallback area under the normalized anytime-regret curve; secondary endpoints are acceptance, empirical one-sided coverage, joint and conditional harm, family effects, and the complete offline/deployment resource ledger. The interval is a frozen 4,000-resample percentile bootstrap within family-size strata, while the conjunctive go/no-go rule also requires portfolio opportunity, negative interval endpoints, at least ten accepted graphs, coverage at least 0.90, and joint harm at most 0.10. The simulator uses exact complex-128 statevectors: there are no shots, device noise, compilation, queue latency, or hardware claims. Runs use Python 3.13.12, NumPy 2.4.3, SciPy 1.17.1, NetworkX 3.6.1, PyTorch 2.11.0, Matplotlib 3.10.8, and pandas 3.0.2 on CPU, with master seed 20260719. From the repository root, `python run.py release` regenerates validated displays, checks source synchronization and evidence hashes, builds the manuscript and source archive, and fails on drift; `gctr-reproduce-all replay` and its `full -quick` mode provide installed-package replay and reduced-cost execution.

A. Development Selection

Graphs come from four synthetic families: Erdős–Rényi [56], random regular as generated by NetworkX [57], Barabási–Albert [58], and Watts–Strogatz [59]. Sequential filtering rejects isomorphic repeats within each stratum, and separate angle-label, development, calibration, and audit splits are frozen by graph identity. An initial fixed-parameter generator grid used all four families without broad parameter variation; a subsequent 64-graph heterogeneous development audit used declared ranges and triggered the family-fallback protocol. The locked selector was then fit on 48 development graphs and its residual threshold computed from 48 disjoint calibration graphs, all before the confirmatory list and decision were frozen.

For any split of N graphs, let $\bar{u}_{\mathrm{global}}$ be the mean utility of the single global arm and $\bar{u}_{\mathrm{oracle}} = N^{-1} \sum_i \min_a u_{ia}$ the mean per-graph portfolio oracle. Portfolio opportunity is

$$\mathcal{O} = \frac{\bar{u}_{\mathrm{global}} - \bar{u}_{\mathrm{oracle}}}{\bar{u}_{\mathrm{global}}}. \tag{13}$$

The heterogeneous development audit recorded $\mathcal{O} =$

0.10710; the locked confirmatory audit retained the requirement $\mathcal{O} \geq 0.05$ and recorded $\mathcal{O} = 0.07115$.

The disjoint split roles and their exact-query counts are reported in Table I, while Fig. 4 records arm opportunity and split stability.

B. Confirmatory Audit

Table II gives the locked policy comparison, and Fig. 5 displays the frozen residual rule and its inferential boundary. The one-shot audit contains 160 graphs, 40 per family. Gated deployment has mean utility 0.16930 against 0.17207 for the family fallback and 0.17277 for the global development arm, with per-graph oracle 0.16048; the selected-minus-family difference is -0.002768, and a frozen 4,000-resample stratified percentile bootstrap interval is $[-0.005696, -0.00008178]$. That interval is not the decision. The policy accepts 12 routes, incurs 4 joint harms, and has empirical residual coverage 0.88125 against nominal 0.90; since coverage is below the internally frozen threshold, the reliability gate fails. A binomial sensitivity model gives probability 0.249 of observing at most 141 covered cases under independent 0.90 coverage, so the miss is compatible with nominal variation yet still fails the literal frozen rule. Family-conditional gains are heterogeneous: the improvement is largest for random regular graphs (-0.013269) while Barabási–Albert graphs regress ($+0.003764$) and Watts–Strogatz graphs move slightly (-0.001567).

C. Anytime Behavior and Resource Accounting

Figures 6, 7, and 8 separate selective routing error, checkpoint behavior, and family-conditional effects. Every arm receives 32 exact objective calls, so the audit consumes 288 calls per graph across nine arms and 46,080 calls for the 160 audit graphs; the angle-label track is reported separately, and runtime is descriptive because Python and simulator overhead differ by arm. The simulator uses PyTorch for numerical kernels [60] and includes no shot noise, gate noise, compilation, or queue latency. Analytic gradients and higher derivatives can be estimated on hardware under appropriate gates [61], and quantum-natural-gradient methods alter the local metric [62]; neither is evaluated here. The audit's statistical unit is the graph, and the bootstrap resamples within the eight family-by-size strata [52], so the fixed quota improves balance while leaving the pooled sample non-iid.

VII. CONCLUSION

Graph structure carries useful information about QAOA optimizer behavior, but useful information is not the same as reliable routing. The frozen selector slightly improves mean anytime regret over a strong family fallback, yet its accepted set is small, one third of accepted routes are harmful, and the internally frozen empirical-coverage requirement fails, so we reject the confirmatory reliability claim. The value of the study is methodological: target-free anytime utility, matched controls, split-and-lock chronology, and a report that keeps coverage, acceptance, joint harm, and conditional harm distinct.

LIMITATIONS

The experiment uses exact statevectors, depth two, small unweighted MaxCut graphs, fixed graph-family quotas, a known family label at deployment, and a hand-designed nine-arm portfolio, and it provides no hardware, noise, asymptotic scaling, or quantum-advantage result. The percentile bootstrap endpoint is boundary sensitive, and the pooled exchangeability condition required by the simplest order-statistic theorem is not established. A future experiment should replace the descriptive gate with a prospective sampling design, predefine the graph population and filtering procedure, increase accepted-route power, add shot and device noise, and compare against stronger adaptive controls; if several reliability criteria are tested, their multiplicity correction and joint decision rule should be fixed before analysis [53], and any exact binomial or concentration bound must state its independence or exchangeability assumptions [34, 63].

DATA AVAILABILITY

All graph instances, locked portfolio traces, split summaries, and machine-readable decision records used by the manuscript are included in the public repository at `https://github.com/thmolena/QAOA-Graph-Conditioned-Trust-Regions`. The artifact manifest binds every displayed numerical file to a SHA-256 digest and records the frozen protocol, implementation, configuration, and decision identities. No external or restricted dataset is required, and no hardware data were collected.

CODE AVAILABILITY

The repository contains the complete study implementation, frozen configurations, deterministic artifact generator, validators, tests, and source-archive builder. Installing the repository with `pip install .` exposes `gctr-optimize`, `gctr-portfolio`, and `gctr-reproduce-all`. The wheel contains its evidence bundle and supports immutable replay without the original checkout. Exact-statevector calls are reported as such and are not represented as hardware shots.

TABLE I. **Frozen data roles and exact-query counts.** Each role uses a disjoint graph split.

Split	Per family-size	Graphs	Exact queries	Role
Angle labels	6	48	5,760	legacy-arm training
Development	6	48	13,824	routing and fallback
Calibration	6	48	13,824	residual order statistic
Confirmatory audit	20	160	46,080	one-shot decision

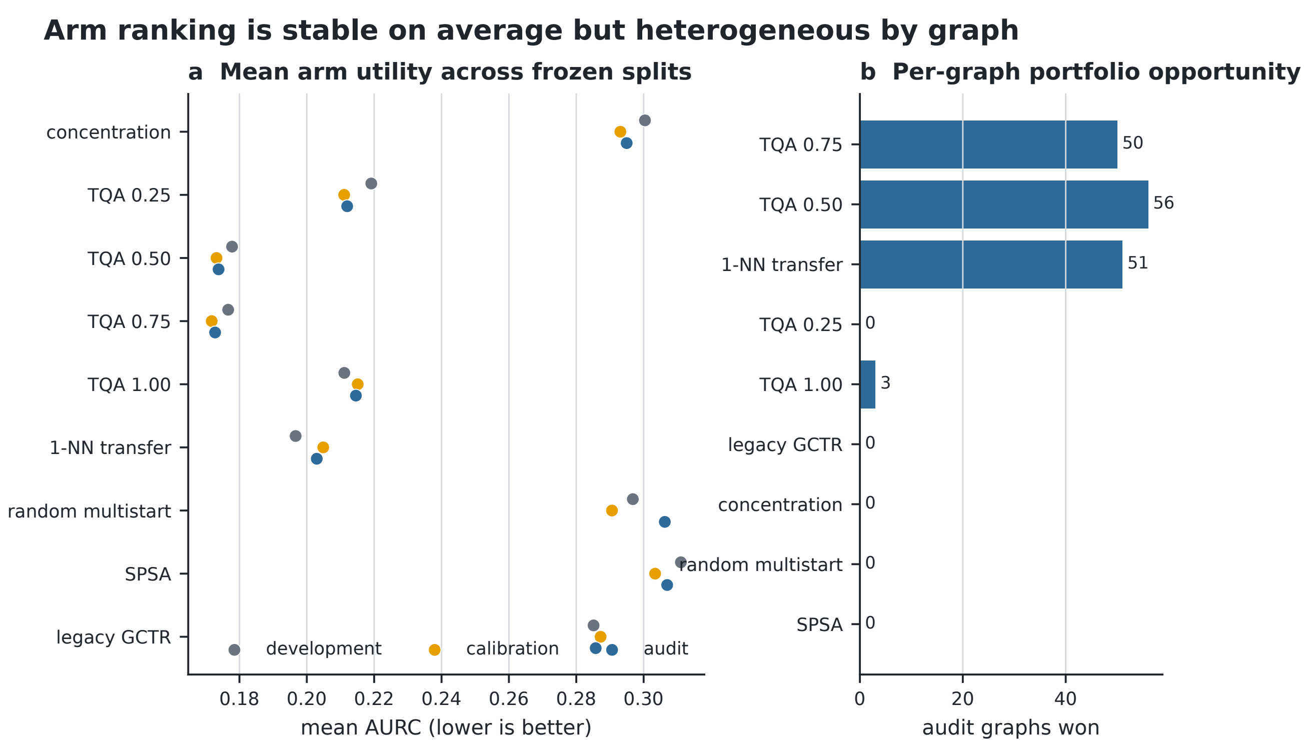

FIG. 4. **Arm opportunity and split stability.** **(a)** Mean utility for every arm on the frozen development, calibration, and audit splits; lower is better. **(b)** Number of audit graphs on which each arm is best. Heterogeneous graph-level winners motivate a portfolio but do not by themselves validate a learned router.

TABLE II. **Confirmatory results and policy comparison.** **(a)** Family-stratified audit results, where lower utility is better and Δ is gated minus family fallback. **(b)** Locked policies on the same 160 audit graphs; the oracle is retrospective and unavailable at deployment.

(a) Family-stratified confirmatory audit.

Stratum	N	Fallback	Gated	Δ	Accepted	Harms
Erdős–Rényi	40	0.16439	0.16439	+0.00000	0	0
random regular	40	0.17223	0.15896	-0.01327	5	0
Barabási–Albert	40	0.18775	0.19151	+0.00376	6	4
Watts–Strogatz	40	0.16391	0.16234	-0.00157	1	0
All	160	0.17207	0.16930	-0.00277	12	4

(b) Locked policy comparison.

Policy	Mean AURC	Δ versus family	Interpretation
Global development arm	0.17277	+0.00070	deployable control
Family fallback	0.17207	+0.00000	deployable control
Gated selector	0.16930	-0.00277	12 accepted routes
Per-graph portfolio oracle	0.16048	-0.01159	retrospective only

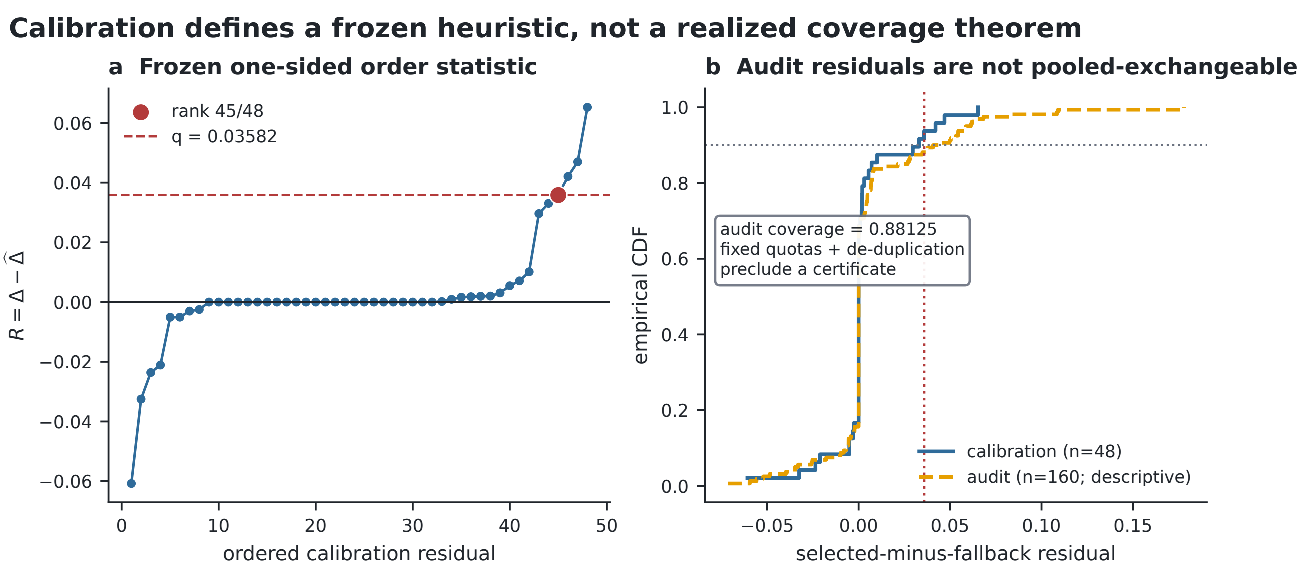

FIG. 5. **Frozen residual rule and its inferential boundary.** **(a)** The one-sided calibration statistic is the rank-45 order statistic among 48 residuals. **(b)** Calibration and audit empirical CDFs are shown descriptively. Fixed family quotas and sequential de-duplication do not justify a pooled exchangeability certificate.

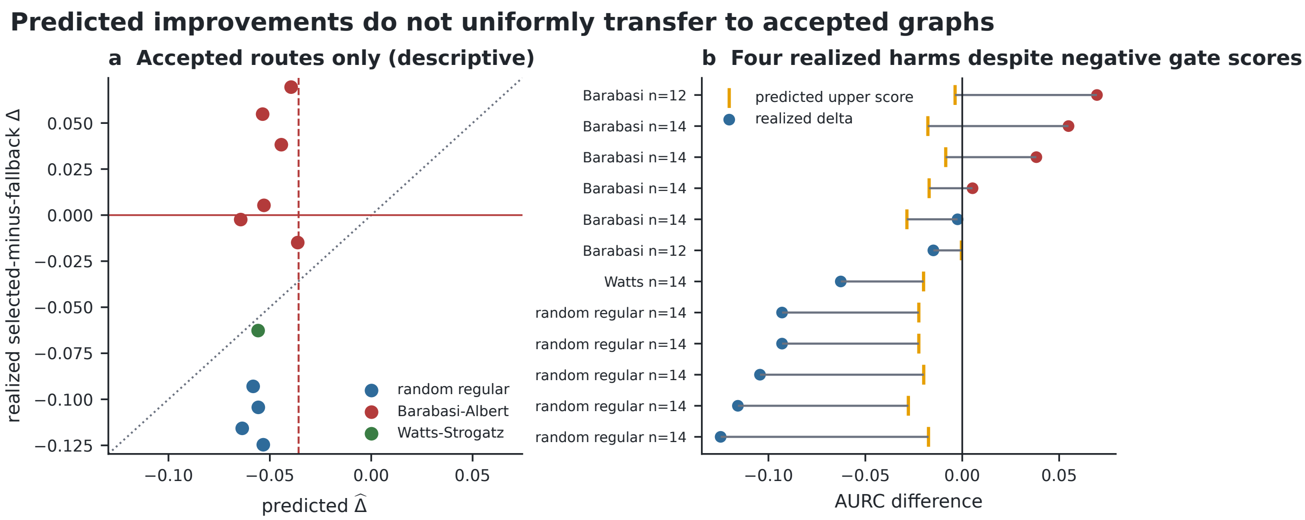

FIG. 6. **Predicted versus realized accepted routes.** **(a)** Predicted and realized selected-minus-fallback differences for the 12 accepted graphs. **(b)** Four accepted routes are harmful despite negative frozen upper scores. The display diagnoses selective failure; it does not refit the gate.

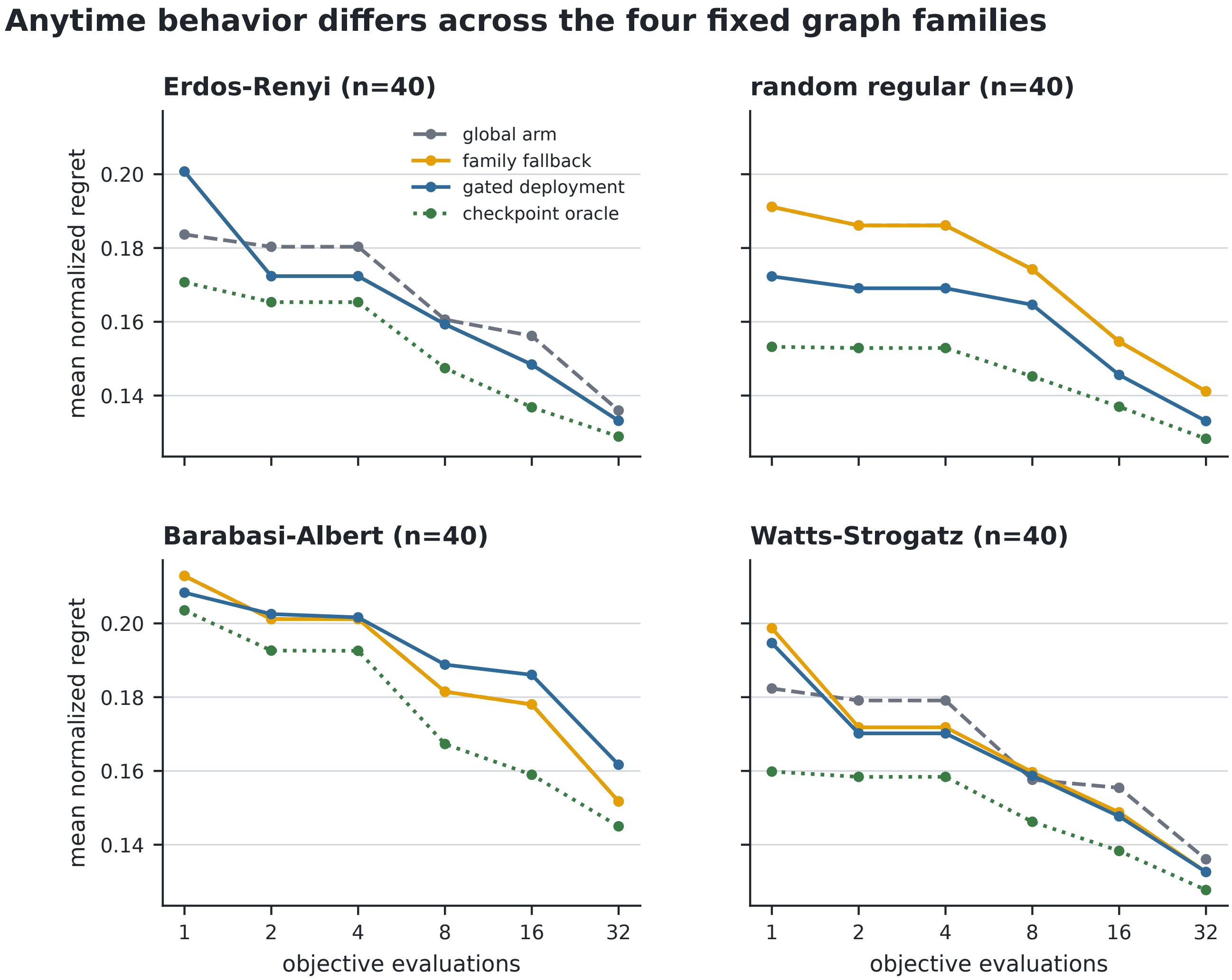

FIG. 7. **Anytime behavior by fixed graph family.** Mean normalized regret is shown at every recorded checkpoint for the global arm, family fallback, gated deployment, and per-checkpoint oracle. Family heterogeneity persists across the 32-call budget.

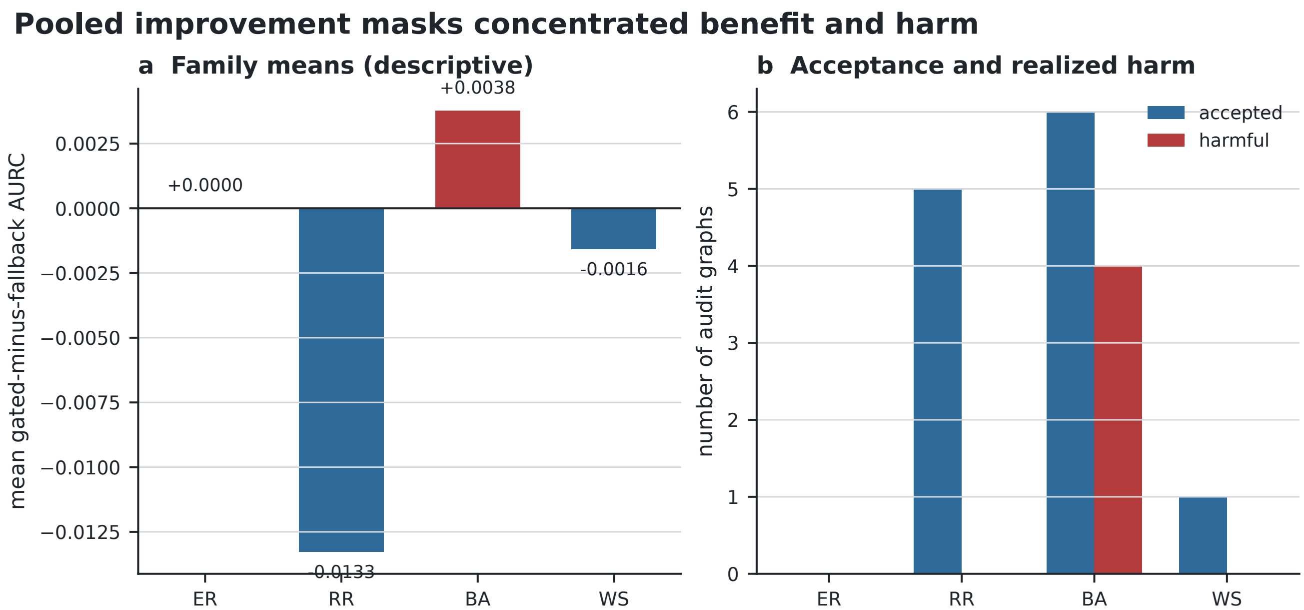

FIG. 8. **Family-conditional effects.** **(a)** Mean gated-minus-fallback utility within each fixed family. **(b)** Acceptance and realized-harm counts. The pooled improvement is concentrated in random regular graphs, whereas all four observed harms occur in the Barabási–Albert stratum.

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Appendix A: Proofs

1. Proof of Theorem IV.1

Proof. Let the instance space contain two points x_0 and x_1, and define losses

$$\ell(a, x_0) = 0, \quad \ell(b, x_0) = \varepsilon,$$
$$\ell(a, x_1) = 1, \quad \ell(b, x_1) = \varepsilon,$$

each of which lies in $[0, 1]$ because $\varepsilon \in (0, 1)$. Let P_0 place probability one on x_0 and P_1 probability one on x_1. Under P_0, $\mathbb{E}_{P_0}\ell(a, X) = 0 < \varepsilon = \mathbb{E}_{P_0}\ell(b, X)$, so a is the unique development risk minimizer; under P_1, $\mathbb{E}_{P_1}\ell(b, X) = \varepsilon < 1 = \mathbb{E}_{P_1}\ell(a, X)$, so b is the unique deployment risk minimizer. A router that unconditionally deploys the development minimizer a has deployment excess risk

$$\mathbb{E}_{P_1}\ell(a, X) - \min_{c\in\{a,b\}} \mathbb{E}_{P_1}\ell(c, X) = 1 - \varepsilon.$$

Because ε can be taken arbitrarily close to zero, this excess approaches the largest possible gap for losses in $[0, 1]$, and no estimation error or model misspecification is needed to produce it. □

2. Proof of Proposition IV.2

Proof. By the definition of conditional probability, when $\mathbb{P}(A) > 0$ we have $\mathbb{P}(H \mid A) = \mathbb{P}(A \cap H)/\mathbb{P}(A)$, and multiplying both sides by $\mathbb{P}(A)$ gives Eq. (8). For the second claim, fix any joint-harm ceiling $\eta \in (0, 1)$ and any $c \in [\eta, 1]$, and construct events with $\mathbb{P}(A) = \eta/c$ and $\mathbb{P}(H \mid A) = c$; then $\mathbb{P}(A \cap H) = \mathbb{P}(A)\mathbb{P}(H \mid A) = \eta$ while the conditional harm equals c. Taking $c = 1$ and $\mathbb{P}(A) = \eta$ in particular yields joint harm η with every accepted route harmful. Thus a bound on $\mathbb{P}(A \cap H)$ alone cannot imply any nontrivial bound on $\mathbb{P}(H \mid A)$ unless acceptance is bounded away from zero, which is the assertion of the proposition. □

3. Proof of Proposition IV.3

Proof. Take any outcome in $A \cap H$. Membership in A gives $\widehat{d} + q < 0$, hence $-\widehat{d} > q$, and membership in H gives $\Delta > 0$. Therefore

$$R = \Delta - \widehat{d} > -\widehat{d} > q,$$

so the outcome lies in $\{R > q\}$; since the outcome was arbitrary, $A \cap H \subseteq \{R > q\}$, which is Eq. (9). Taking probabilities of both sides of the inclusion gives $\mathbb{P}(A \cap H) \le \mathbb{P}(R > q) = 1 - \mathbb{P}(R \le q)$. If $\mathbb{P}(R \le q) \ge 1 - \alpha$ then the right-hand side is at most α, so $\mathbb{P}(A \cap H) \le \alpha$; and when $\mathbb{P}(A) > 0$, dividing by $\mathbb{P}(A)$ gives $\mathbb{P}(H \mid A) \le \alpha/\mathbb{P}(A)$. Each step follows from the deterministic containment and the definition of conditional probability, and no exchangeability assumption enters this argument; exchangeability is relevant only to establishing the separate premise $\mathbb{P}(R \le q) \ge 1 - \alpha$ from calibration data. □

4. Proof of Proposition IV.4

Proof. Uniformly randomizing the audit order is equivalent to choosing the n_f positions occupied by family f uniformly among the $\binom{n}{n_f}$ subsets of size n_f. For any position i,

$$\mathbb{P}(Z_i = 1) = \frac{\binom{n-1}{n_f-1}}{\binom{n}{n_f}} = \frac{n_f}{n} = p_f,$$

and for distinct positions $i \ne j$,

$$\mathbb{P}(Z_i = 1, Z_j = 1) = \frac{\binom{n-2}{n_f-2}}{\binom{n}{n_f}} = \frac{n_f(n_f - 1)}{n(n-1)},$$

with the convention $\binom{N}{k} = 0$ for $k < 0$, which also covers $n_f = 1$. Therefore

$$\begin{aligned}\mathrm{Cov}(Z_i, Z_j) &= \frac{n_f(n_f - 1)}{n(n-1)} - \frac{n_f^2}{n^2} \\ &= -\frac{n_f(n - n_f)}{n^2(n-1)} = -\frac{p_f(1 - p_f)}{n-1},\end{aligned}$$

which is strictly negative because $0 < n_f < n$. Independent Bernoulli variables have zero covariance, so the family indicators cannot be independent; their joint law is invariant under permutations of positions, hence exchangeable. Finally, iid sampling from a mixture with family probability p_f would give a binomial family count with variance $np_f(1 - p_f) > 0$, whereas the quota design fixes that count at n_f with variance zero, so the design is not pooled iid mixture sampling. □

5. Proof of Theorem V.1

Proof. Because $R_1, \dots, R_{n+1}$ are exchangeable and almost surely distinct, the rank

$$\mathrm{rank}(R_{n+1}) = 1 + \sum_{i=1}^{n} \mathbf{1}\{R_i < R_{n+1}\}$$

is uniform on $\{1, \dots, n + 1\}$. Let q be the kth smallest calibration score. If $k = n + 1$, the convention $q = +\infty$ gives $\mathbb{P}(R_{n+1} \le q) = 1 = k/(n + 1)$ and the claim holds, so assume $1 \le k \le n$. If the test rank is at most k, then at most $k - 1$ calibration scores are smaller than R_{n+1}, so the kth calibration score is at least R_{n+1} and $R_{n+1} \le q$; if the test rank exceeds k, then at least k calibration scores are smaller and $R_{n+1} > q$. Hence $\{R_{n+1} \le q\} = \{\mathrm{rank}(R_{n+1}) \le k\}$, and uniformity of the rank gives $\mathbb{P}(R_{n+1} \le q) = k/(n + 1)$. Finally $k = \lceil (n+1)(1-\alpha) \rceil \ge (n+1)(1-\alpha)$, so dividing by $n+1$ yields $\mathbb{P}(R_{n+1} \le q) \ge 1 - \alpha$. □

Appendix B: Frozen Structural Routing

Frozen structural-routing procedure. Given a graph G, development standardization, ridge coefficients, a family fallback map, and the order statistic q:

1. compute the 16 relabeling-invariant summaries $x(G)$;
2. standardize them using development means and scales;
3. predict all nine utilities with Eq. (11);
4. choose $s = \arg\min_a \widehat{u}_a(x)$ and retrieve $b(f)$;
5. accept s only if $\widehat{u}_s - \widehat{u}_b + q < 0$; otherwise deploy b;
6. run the deployed arm under a hard 32-call counter and record every best-so-far checkpoint.

The audit implementation additionally verifies that the graph, configuration, decision, and implementation hashes match the frozen manifest before any evaluation. The nine arms combine model-based and model-free search; Gaussian-process optimization is a natural future arm [39, 41], and randomized low-rank models may reduce structural fitting cost on much larger feature sets [64], though neither is needed for the released 16-feature ridge fit.

Appendix C: Risk Quantities and Reliability Sensitivity

Ridge predictions carry estimation error from finite development data, model bias from linearity, and distribution shift across strata, while the selected-minus-fallback residual additionally contains selection-induced dependence because the selected arm is an $\arg\min$ of the same prediction vector; calibration treats the resulting scalar as a conformity score without assuming Gaussianity. Figure 9 pairs empirical coverage-failure, joint-harm, and conditional-harm rates with one-sided exact-binomial upper bounds under a hypothetical iid model: the 141 covered residuals out of 160 fall below the literal 0.90 requirement, and a binomial model with success probability 0.90 places lower-tail probability 0.249 on that count. These are sensitivity analyses, not certificates for the dependent fixed-quota design. Broader operator portfolios could eventually motivate shadow-tomography or classical-shadow measurement [65, 66], but the present commuting MaxCut objective does not require them. The released legacy optimizer-ablation artifact remains a reproducible historical development diagnostic and is kept outside the locked confirmatory portfolio decision.

The released legacy optimizer-ablation artifact remains a reproducible historical development diagnostic. It informs arm design but is excluded from the locked confirmatory decision.

Appendix D: Optimizer Portfolio and Frozen Arms

The concentration arm uses the coordinate-wise median of the 48 angle-label training vectors; four linear schedules interpolate fixed midpoint ramps; one-nearest-neighbor transfer retrieves the closest angle-label training graph in standardized feature space; multistart divides the budget across four seeded starts; SPSA uses a frozen gain schedule; and the legacy trust-region arm maps graph features to a Gaussian angle policy and a local budget rule, receiving no extra calls. The lowest development mean-utility arm within the known generator family is the deployable fallback, and the structural $\arg\min$ replaces it only when Eq. (7) certifies a negative predicted upper bound.

The frozen portfolio and its realized selection-to-deployment transitions are given in Table III and Fig. 10.

Appendix E: Split Diagnostics and Locked Gate

Development and calibration use disjoint graph lists, and feature standardization, ridge fitting, family-arm means, and residual sorting use only their assigned roles; the calibration quantile is the conservative upper order statistic for $\alpha = 0.10$. The frozen gate applies the recorded decision rule literally to the locked summaries and does not reinterpret a failed criterion after audit inspection. The confirmatory configuration fixes 20 graphs per family-size stratum for 160 total, and the decision manifest stores the selected arm, fallback, predicted delta, acceptance flag, and relevant hashes before audit traces are scored. Development-stage figures record 10 accepted routes, 1 joint harm, and coverage 0.9375, with paired difference -0.009370 and interval $[-0.016454, -0.002500]$.

Table IV records the frozen decision, and Fig. 11 reports the corresponding split diagnostics.

Appendix F: Resource Accounting

Every optimizer arm receives exactly 32 objective calls: fixed schedules still pay for their seed evaluation, multistart pays for each start from the same counter, and model training and angle-label generation are reported outside the deployable per-graph budget. The displayed ledger covers the final locked protocol and totals 79,488 exact objective calls; the earlier regular-grid and heterogeneous development-stage studies used 36,096 and 51,840 additional calls, so repeating all three committed evidence tracks would use 167,424 exact objective calls. These are research-construction costs, not the 32-call cost of deploying one frozen arm on a new graph, and exact statevector runtime is never converted into hardware shots.

Figure 12 and Table V report the two accounting views.

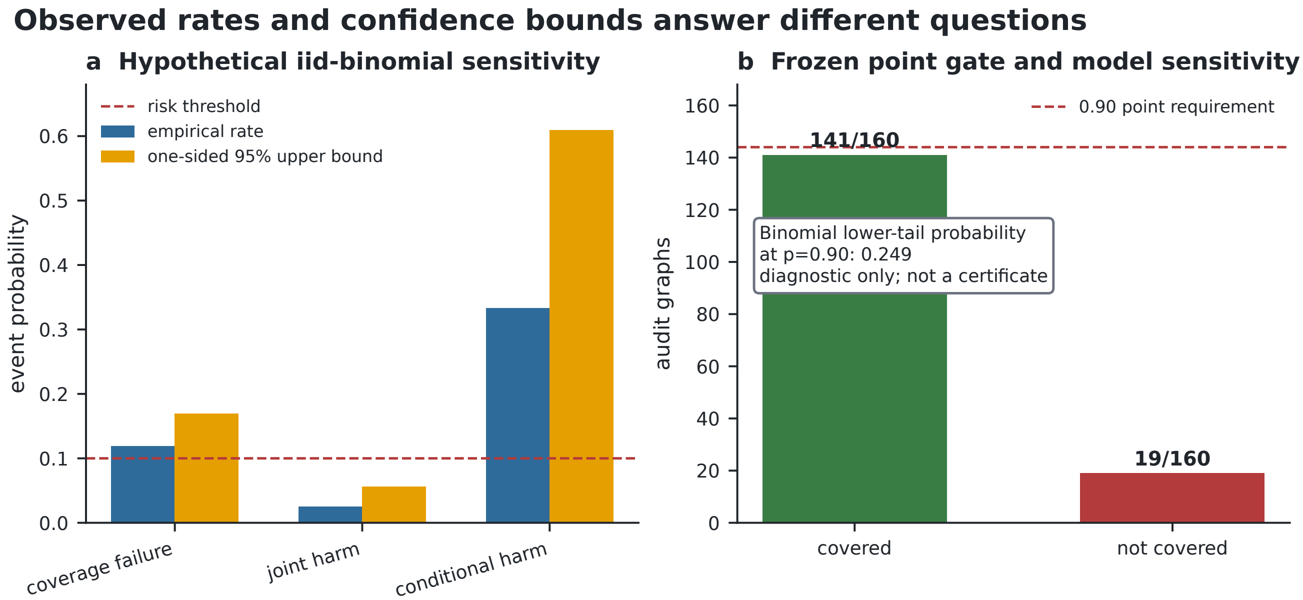

FIG. 9. **Risk quantities are not interchangeable. (a)** Empirical coverage-failure, joint-harm, and conditional-harm rates with one-sided exact-binomial upper bounds under a hypothetical iid model. **(b)** The observed coverage count against the frozen point requirement. These are sensitivity analyses, not certificates for the dependent design.

TABLE III. **Frozen nine-arm portfolio.** Every arm receives the same 32-call ceiling; offline labels and search rules are shown explicitly.

Arm	Initialization	Search rule	Offline labels	Calls
Concentration	coordinate-wise median label	pattern search	yes	32
TQA $d_t = 0.25$	midpoint linear ramp	pattern search	no	32
TQA $d_t = 0.50$	midpoint linear ramp	pattern search	no	32
TQA $d_t = 0.75$	midpoint linear ramp	pattern search	no	32
TQA $d_t = 1.00$	midpoint linear ramp	pattern search	no	32
1-NN transfer	nearest standardized training graph	pattern search	yes	32
Random multistart	four uniform random starts	four pattern searches	no	32
SPSA	concentration angle	$a = 0.18, c = 0.12, \alpha = 0.602, \gamma = 0.101$	yes	32
Legacy GCTR	GIN diagonal-Gaussian mean	radius-two covariance-shaped pattern search	yes	32

The released evidence provenance record binds the protocol, implementation, configuration, and frozen decision to complete SHA-256 values checked before the manuscript assets are accepted. These identities establish which records were analyzed; they do not replace statistical validation.

The post-hoc inferential and threshold diagnostic is shown separately in Fig. 13.

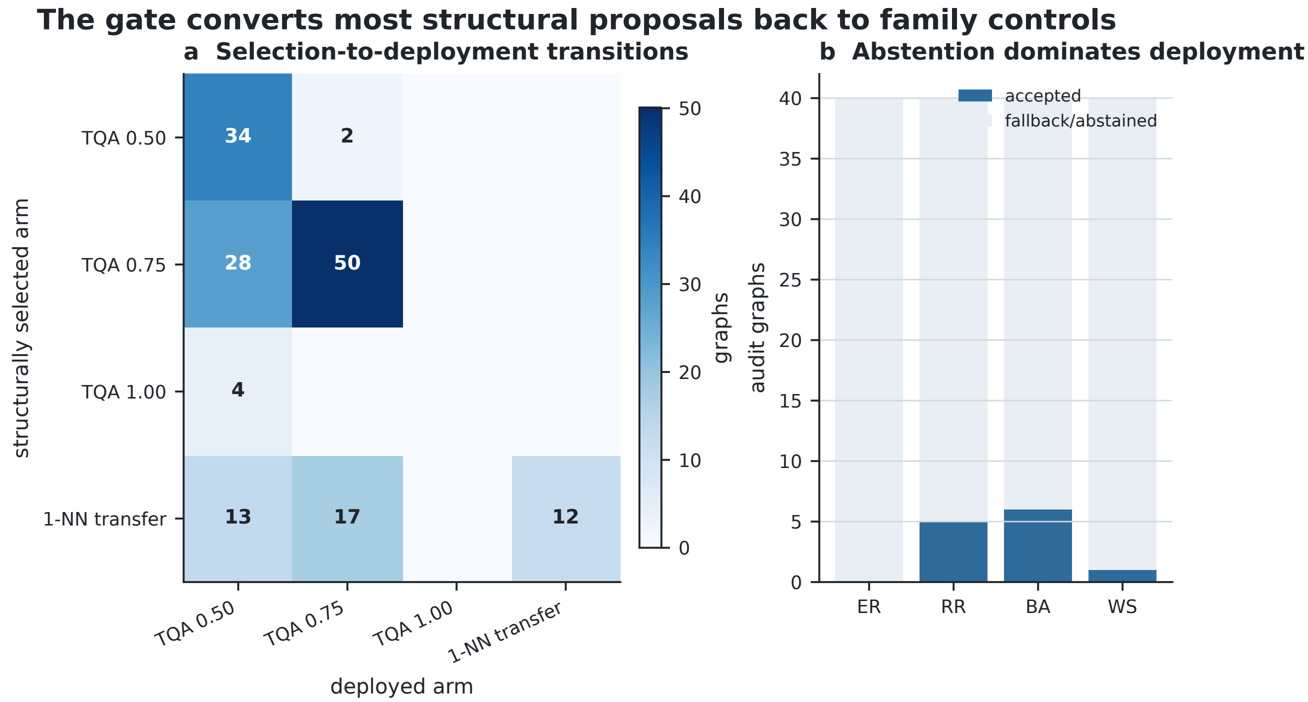

FIG. 10. **Selection-to-deployment transitions.** **(a)** Counts from the structurally selected arm to the actually deployed arm. **(b)** Accepted and abstained graphs by family. The gate returns most structural proposals to their family control.

TABLE IV. **Frozen confirmatory gate.** The empirical-coverage component fails, so the overall decision is no-go.

Criterion	Required	Observed	Result
Portfolio opportunity	≥ 0.05	0.071	pass
Global-baseline interval upper end	< 0	-0.000646	pass
Family-fallback interval upper end	< 0	-0.00008178	pass
Accepted audit graphs	≥ 10	12	pass
One-sided empirical coverage	≥ 0.90	0.88125	**fail**
Joint harm rate	≤ 0.10	0.0250	pass

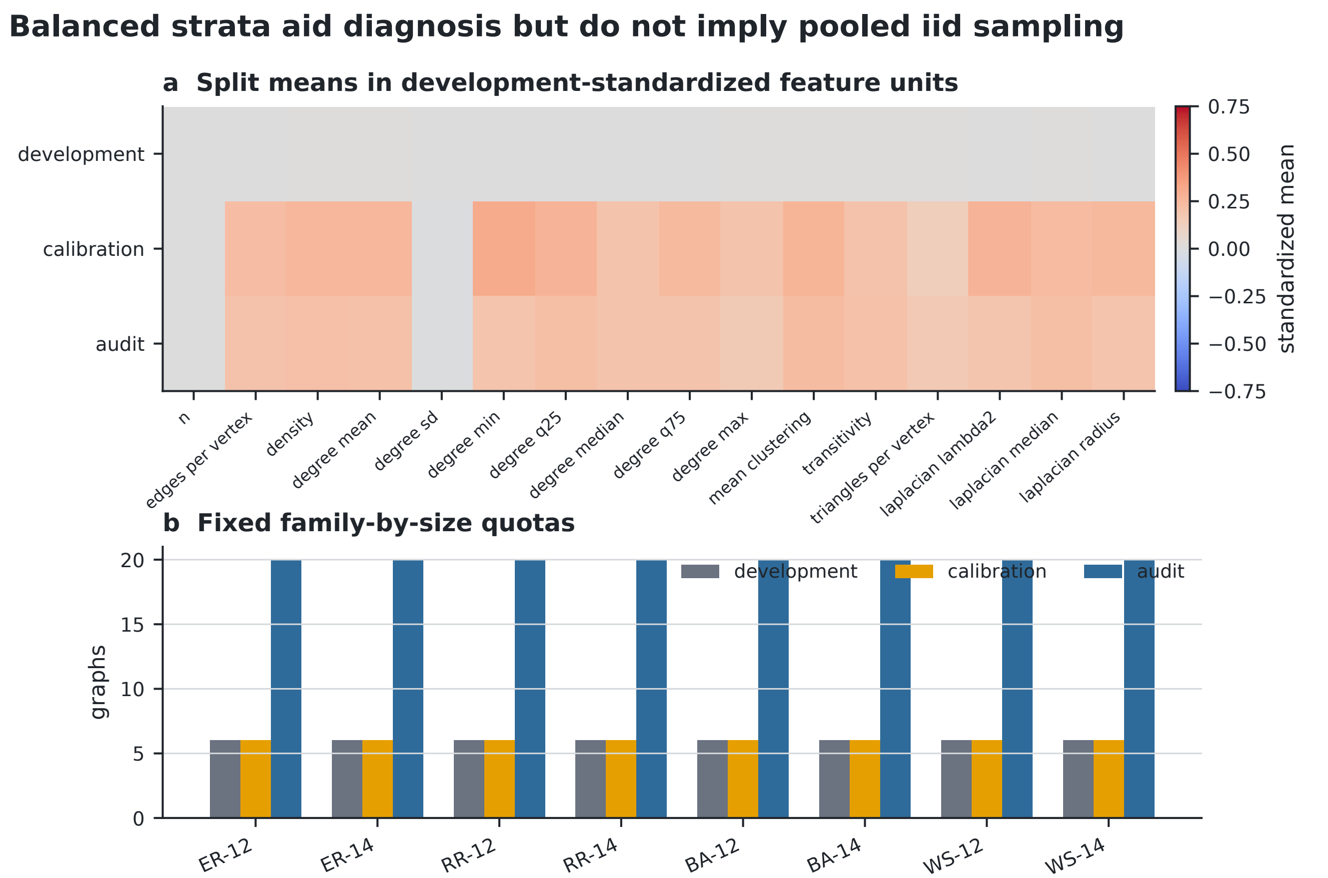

FIG. 11. **Split diagnostics.** **(a)** Feature means in development-standardized units for development, calibration, and audit. **(b)** Fixed family-by-size quotas. Balance aids stratified diagnosis but does not make the pooled sample iid.

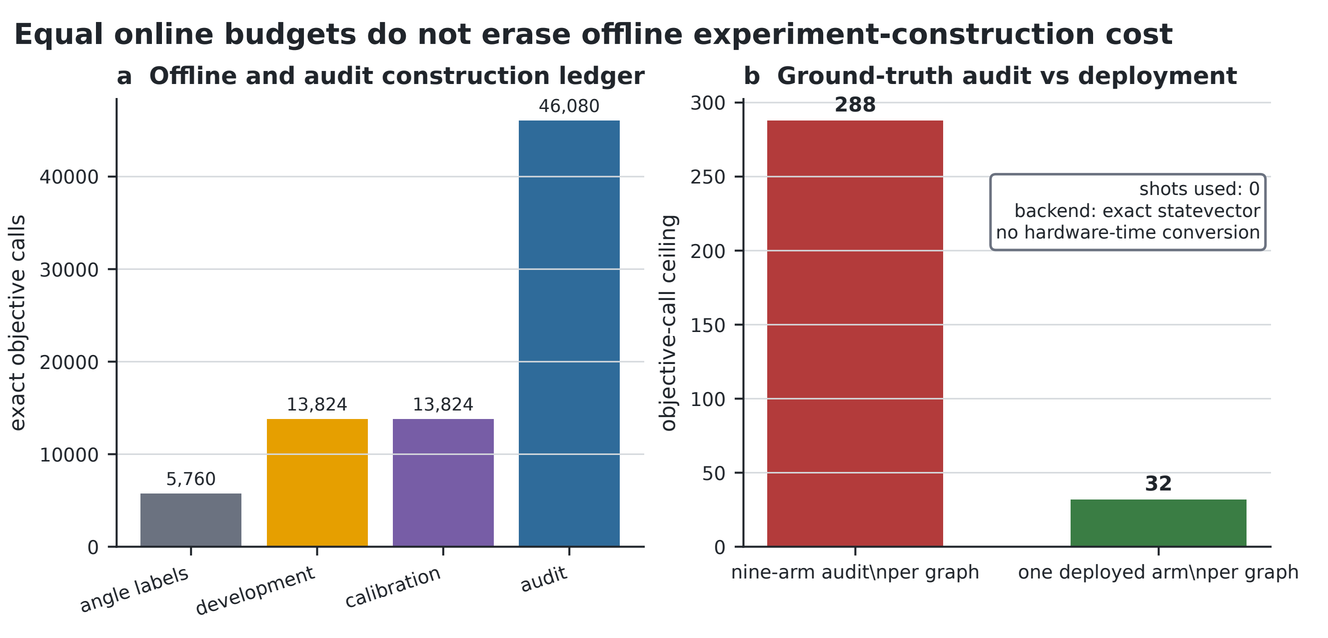

FIG. 12. **Exact-statevector resource ledger.** **(a)** Exact objective calls used for angle labels, development, calibration, and the one-shot audit. **(b)** The nine-arm research audit is separated from the 32-call ceiling of one deployed arm. No conversion to shots or hardware time is made.

TABLE V. **Final-protocol exact-statevector resource ledger.** Query totals are regenerated from the locked confirmatory summary; earlier development-stage studies are reported in the surrounding text, and the deployed per-graph ceiling is not multiplied by the nine-arm research audit.

Stage	Graphs	Exact objective calls	Accounting role
Angle-label training	48	5,760	legacy-arm labels
Development traces	48	13,824	nine-arm portfolio
Calibration traces	48	13,824	nine-arm portfolio
Confirmatory traces	160	46,080	nine-arm audit
Deployed policy	1	32	per new graph

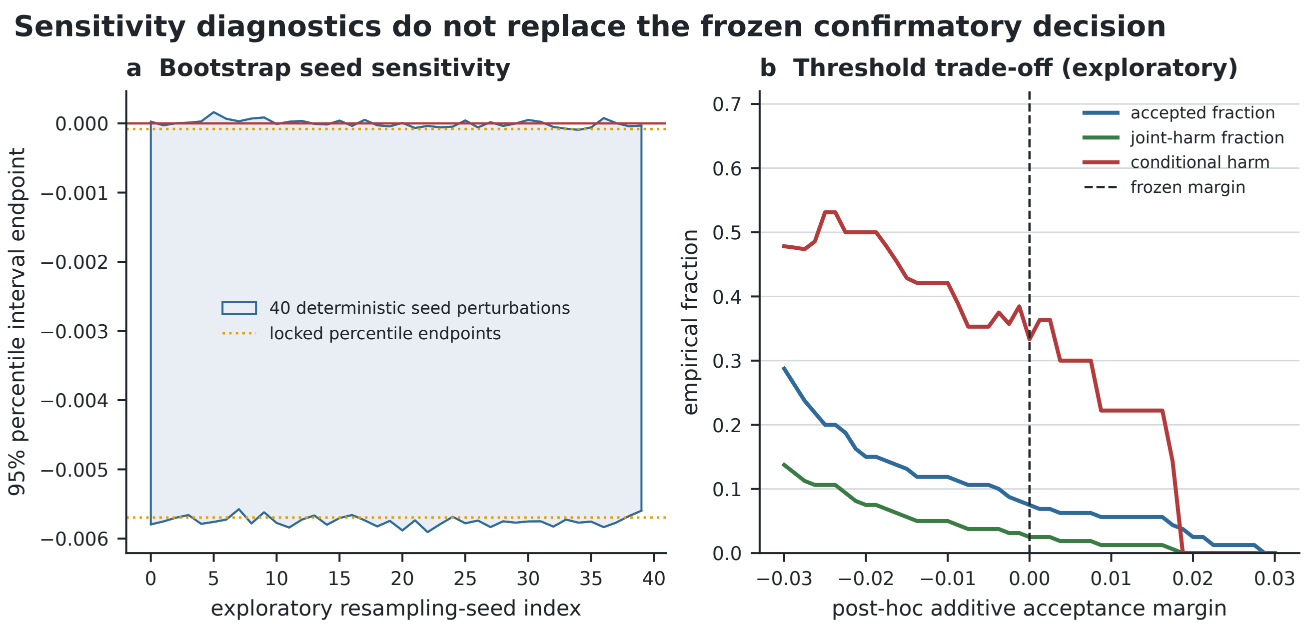

FIG. 13. **Exploratory inference and threshold sensitivity.** **(a)** Percentile-bootstrap endpoints under deterministic resampling-seed perturbations. **(b)** Acceptance and harm rates under post-hoc additive gate margins. Neither panel replaces or modifies the frozen confirmatory decision.